

Tarea 10.3

$$13) 144x^2 - 25y^2 + 864x - 100y - 2404 = 0$$

$$(144x^2 + 864x) - 25y^2 - 100y = 2404$$

$$144(x^2 + 6x) - 25y^2 - 100y = 2404$$

$$144(x^2 + 6x + 9) - 25y^2 - 100y = 3,700$$

$$144(x+3)^2 - 25(y^2 - 4y + 4) = 3,600$$

$$144(x+3)^2 - 25(y+2)^2 = 3,600$$

$$\frac{(x+3)^2}{25} - \frac{(y+2)^2}{144} = 1$$

$$\text{Centro} = (-3, -2)$$

$$a = 5$$

$$b = 12$$

$$c = \sqrt{144 + 25}$$

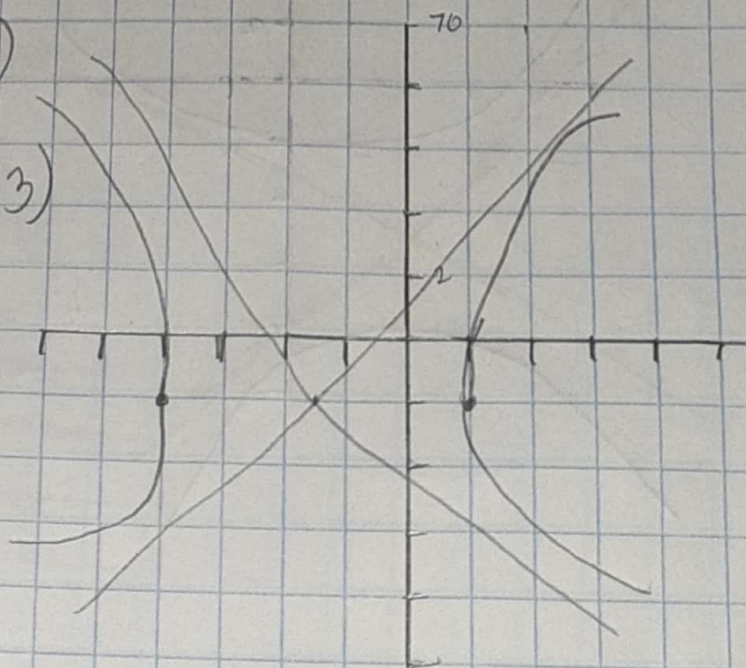
$$c = \sqrt{169} = 13$$

$$V = (2, -2) \text{ y } (-8, -2)$$

$$F = (h \pm c, k)$$

$$= (10, -2) \text{ y } (-16, -2)$$

$$\text{Asint} = (y+2) = \pm \frac{12}{5}(x+3)$$



$$15) 4y^2 - x^2 + 40y - 4x + 60 = 0$$

$$4(y^2 + 10y + 25) - x^2 - 4x = 40$$

$$4(y+5)^2 - (x^2 + 4x + 4) = 40$$

$$4(y+5)^2 - (x+2)^2 = 36$$

$$\frac{(y+5)^2}{9} - \frac{(x+2)^2}{36} = 1$$

$$\text{Centro} = (-2, -5)$$

$$a = 6$$

$$b = 3$$

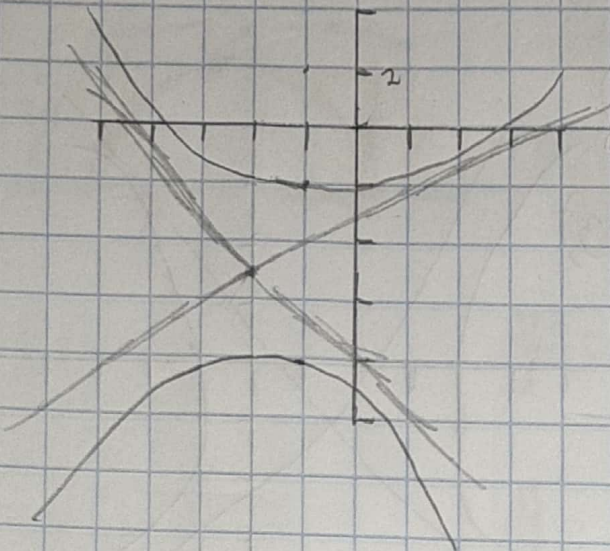
$$c = \sqrt{a^2 + b^2}$$

$$c = 3\sqrt{5}$$

$$V = (-2, -2) \text{ y } (-2, -8)$$

$$F = (-2, -5 + 3\sqrt{5}) \text{ y } (-2, -5 - 3\sqrt{5})$$

$$\text{Directrices} = (y+5) = \pm \frac{3}{6} (x+2) = (y+5) = \pm \frac{1}{2} (x+2)$$



$$25) F(0, \pm 5)$$

$$\text{Centro} = (0, 0)$$

$$\text{eje conjugado} = 4 \quad 4/2 = 2$$

$$\frac{y^2}{25} - \frac{x^2}{4} = 1$$

$$c^2 = a^2 + b^2$$

$$25 = 4 + b^2$$

$$21 = b^2$$

$$37) \frac{1}{3}(x+2) = y^2$$

$$y^2 = \frac{1}{3}(x+2)$$

parábola eje horizontal

$$39) x^2 + 6x - y^2 = 7$$

$$(x^2 + 6x + 9) - y^2 = 16$$

$$(x+3)^2 - y^2 = 16$$

$$\frac{(x+3)^2}{16} - \frac{y^2}{16} = 1 \quad \text{hipérbola}$$

$$41) -x^2 = y^2 - 25$$

$$-x^2 - y^2 = -25$$

$$x^2 + y^2 = 25 \quad \text{círculo}$$

$$43) 4x^2 - 76x + 9y^2 + 36y = -16$$

$$4(x^2 - 19x) + 9y^2 + 36y = -16$$

$$4(x^2 - 19x + 9) + 9(y^2 + 4y) = 0$$

$$4(x-2)^2 + 9(y^2 + 4y + 4) = 36$$

$$4(x-2)^2 + 9(y+2)^2 = 36$$

$$\frac{(x-2)^2}{9} + \frac{(y+2)^2}{4} = 1 \quad \text{Elipse}$$

$$59) \text{ Rama izquierda de } \frac{x^2}{4} - \frac{y^2}{16} = 1$$

$$\frac{x^2}{4} = 1 + \frac{y^2}{16} \Rightarrow x = -\frac{1}{2} \sqrt{y^2 + 16}$$

$$75) \frac{x^2}{a^2} - \frac{y^2}{b^2} = \frac{50^2}{24^2} - \frac{84^2}{b^2} = 4.34 - \frac{7056}{b^2} = 1$$

$$b^2 = \frac{7056}{4.34} = b = \sqrt{2112}$$

$$c^2 = a^2 + b^2$$

$$\frac{d^2}{24^2} - \frac{36^2}{45.96^2} = 1$$

$$c = \sqrt{576 + 2112.32}$$

$$c = \sqrt{2688.32}$$

$$d = \sqrt{(1.613)(576)}$$

$$d = \sqrt{929.38} = 30.48 \times 2 = 60.97$$

60.97 metros